

Multirate Stein Variational Gradient Descent for Efficient Bayesian Sampling

Anonymous Author(s)

Anonymous Institution

Abstract

Many particle-based Bayesian inference methods use a single global step size for all parts of the update. In Stein variational gradient descent (SVGD), however, each update combines two qualitatively different effects: attraction toward high-posterior regions and repulsion that preserves particle diversity. These effects can evolve at different rates, especially in high-dimensional, anisotropic, or hierarchical posteriors, so one step size can be unstable in some regions and inefficient in others. We derive a multirate version of SVGD that updates these components on different time scales. The framework yields practical algorithms, including a symmetric split method, a fixed multirate method (MR-SVGD), and an adaptive multirate method (Adapt-MR-SVGD) with local error control. We evaluate the methods in a broad and rigorous benchmark suite covering six problem families: a 50D Gaussian target, multiple 2D synthetic targets, UCI Bayesian logistic regression, multimodal Gaussian mixtures, Bayesian neural networks, and large-scale hierarchical logistic regression. Evaluation includes posterior-matching metrics, predictive performance, calibration quality, mixing, and explicit computational cost accounting. Across these six benchmark families, multirate SVGD variants

improve robustness and quality-cost tradeoffs relative to vanilla SVGD. The strongest gains appear on stiff hierarchical, strongly anisotropic, and multimodal targets, where adaptive multirate SVGD is usually the strongest variant and fixed multirate SVGD provides a simpler robust alternative at lower cost.

Keywords: Stein variational gradient descent, Bayesian computation, multirate integration, particle-based variational inference, adaptive time stepping

1. Introduction

Modern Bayesian inference often requires efficient sampling from complex, high-dimensional, and multimodal posterior distributions. As these problems grow in scale and complexity, traditional sampling methods face significant computational and statistical challenges. The advent of automatic differentiation has enabled the widespread use of gradient-based samplers, making methods such as stochastic gradient Langevin dynamics (SGLD) and stochastic gradient Hamiltonian Monte Carlo (SGHMC) practical and popular choices for large-scale applications [1, 2]. While these approaches advance a single Markov chain using noisy gradients, an alternative paradigm is offered by interacting particle methods, which evolve a population of particles whose empirical distribution approximates the target posterior. This population-based view is central to particle-based variational inference [3]. Within this class, SVGD is a prominent deterministic representative, and recent work has examined acceleration, particle degeneracy, and kernel design in related Stein-particle methods [3, 4, 5]. It opens new avenues for improv-

17 ing sampling efficiency and robustness, especially in challenging inference
18 settings.

19 A useful unifying perspective is to view interacting particle methods as
20 discretizations of a particle flow (possibly with a stochastic component) that
21 transports an initial ensemble toward the target distribution [6]. Here, we
22 focus on Stein variational gradient descent (SVGD) [7] as a representative in-
23 teracting particle method. SVGD constructs a velocity field for the particles
24 based on the target density and a reproducing kernel Hilbert space (RKHS)
25 structure, and updates the particles by discretizing this flow [7]. The veloc-
26 ity field naturally separates into an attractive drift, which moves particles
27 toward high-density regions, and a repulsive interaction, which maintains
28 particle diversity and support coverage of multiple modes.

29 This drift-repulsion structure creates the main numerical challenge stud-
30 ied in this paper. When particles crowd, repulsion can become stiff and
31 destabilize updates. On the other hand, the drift can require finer resolution on
32 anisotropic targets to follow the geometry of $\log p$ closely. A single step size
33 must therefore balance competing requirements and can either over-damp
34 repulsion, leading to loss of diversity and mode coverage, or under-resolve
35 drift, leading to slow progress.

36 Moreover, repulsion typically requires kernel evaluation at each step, so
37 increasing the stability by reducing the step-size can become computationally
38 expensive. Finally, repulsive forces can weaken in high dimensions or under
39 limited particle resolution, leading to particle collapse and poor support of
40 multiple modes [4, 8]. In practice, the resulting single-step-size compromise
41 is both statistically and computationally inefficient: it can degrade support

42 coverage and predictive quality while still spending substantial kernel work
43 to stabilize the wrong part of the update.

44 A common approach to solving systems with significantly different com-
45 ponents is to use partitioned methods such as implicit-explicit (IMEX) [9, 10]
46 and multirate integration methods [11]. In the context of ordinary differen-
47 tial equations (ODEs), multirate methods exploit a natural splitting of the
48 dynamics into fast and slow components, allowing each to be integrated with
49 a step size appropriate to its scale [12, 13, 14]. For example, in a split ODE
50 of the form $\dot{x} = \phi_{\text{fast}}(x) + \phi_{\text{slow}}(x)$, multirate schemes apply more frequent,
51 smaller steps to the stiff or rapidly varying part, while updating the slow part
52 less often. This approach improves both stability and efficiency in multi-scale
53 dynamical systems, and has been successfully applied in a variety of scientific
54 computing contexts [10, 15].

55 Motivated by advances in multirate time-integration for multi-scale dy-
56 namical systems—including MrGARK, IMEX, and related splitting schemes
57 [12, 13, 14] we propose to integrate the drift and repulsion terms on sepa-
58 rate time scales. This leads to a number of multirate SVGD variants, in-
59 cluding symmetric splitting, fixed repulsion substepping, and adaptive error-
60 controlled drift substepping.

61 To assess the effectiveness of these methods, we conduct a comprehensive
62 empirical evaluation. We compare our multirate SVGD variants with estab-
63 lished approaches from the literature across a range of benchmark problems,
64 using a variety of task-appropriate metrics. Results highlight the strengths
65 and limitations of each method under diverse geometric and statistical chal-
66 lenges.

67 *Contributions*

68 The main contributions of this work are:

- 69 • Multirate SVGD formulations that explicitly separate drift and repul-
70 sion and support symmetric splitting and fixed multirate schedules.
- 71 • An adaptive error-controlled multirate SVGD variant that selects sub-
72 steps based on a local error estimator while keeping repulsion stable.
- 73 • A broad empirical study showing that multirate SVGD improves ro-
74 bustness and quality-cost tradeoffs compared to SVGD and represen-
75 tative chain methods, with the strongest gains on anisotropic, multi-
76 modal, and hierarchical targets.
- 77 • A benchmark suite spanning many different targets, equipped with
78 appropriate diagnostics, early stopping, performance evaluation metrics
79 for efficiency comparisons.

80 We present our work as follows. Section 2 presents the SVGD variants
81 and the multirate construction. Section 3 describes the benchmarks, diag-
82 nostics, and experimental protocol. Section 4 reports the results. Section 5
83 summarizes the findings and discusses directions for future work.

84 **2. Method**

85 *2.1. Background: SVGD as a particle flow*

86 Let $p(x)$ denote a target density on $\mathbb{R}^d$ and let $\{x_i^t\}_{i=1}^N$ be a set of par-
87 ticles at iteration t . SVGD evolves particles by following a deterministic

88 interacting-particle flow whose velocity field is constrained to RKHS [7]. A
 89 first-order discretization of this flow is the update

$$x_i^{t+1} = x_i^t + h \phi(x_i^t), \quad (1)$$

90 where $h > 0$ is a step size and the SVGD velocity field is

$$\phi(x) = \frac{1}{N} \sum_{j=1}^N \left[k(x_j, x) \nabla_{x_j} \log p(x_j) + \nabla_{x_j} k(x_j, x) \right]. \quad (2)$$

91 The first term in eq. (2) is an *attractive* drift that pushes particles toward
 92 high-density regions via the score $\nabla \log p$. The second term is a *repulsive*
 93 interaction that discourages particle collapse. The multirate constructions
 94 below are kernel-agnostic: they apply once a kernel-induced attractive field
 95 and repulsive field have been specified.

96 2.2. Drift–repulsion splitting and multirate integration

97 The central numerical observation motivating this work is that the drift
 98 and repulsion components of eq. (2) can impose different stability constraints
 99 and can naturally evolve on different time scales. We therefore write the
 100 SVGD flow as a split system

$$\dot{x} = f_{\text{rep}}(x) + f_{\text{drift}}(x), \quad (3)$$

101 where, for particle i ,

$$f_{\text{drift}}(x_i) := \frac{1}{N} \sum_{j=1}^N k(x_j, x_i) \nabla_{x_j} \log p(x_j), \quad (4)$$

$$f_{\text{rep}}(x_i) := \frac{1}{N} \sum_{j=1}^N \nabla_{x_j} k(x_j, x_i). \quad (5)$$

102 We next define multirate discretizations that apply different substepping
 103 strategies to f_{drift} and f_{rep} while keeping a common macro-step size h .

104 *Strang-split SVGD*. Define the forward Euler update that advance the split
 105 system by a step size τ :

$$\Psi_{\text{rep}}^{\tau}(x) = x + \tau f_{\text{rep}}(x), \quad \Psi_{\text{drift}}^{\tau}(x) = x + \tau f_{\text{drift}}(x).$$

106 *Strang splitting* [16] is a classical second-order method that constructs a
 107 macro-step of size h by composing such substeps. In our setting, one macro
 108 step takes the form

$$x^{t+1} = \Psi_{\text{rep}}^{h/2} \circ \Psi_{\text{drift}}^h \circ \Psi_{\text{rep}}^{h/2}(x^t), \quad (6)$$

109 which alternates half a repulsion step, a full drift step, and another half
 110 repulsion step. In practice, the half-steps can be implemented as $\frac{h}{2m}$ smaller
 111 substeps for stability.

112 *Fixed multirate SVGD (MR-SVGd)*. Fixed multirate integrators provide a
 113 principled way to allocate different step sizes to fast and slow components of
 114 a split system [11, 12, 13]. In our SVGD setting, however, higher-order multi-
 115 rate schemes are often not cost-effective: their additional stages translate into
 116 multiple evaluations of the drift and repulsion fields per macro step, and each
 117 such evaluation is expensive due to gradient computations and, in particular,
 118 $O(N^2)$ kernel interactions. Moreover, practical accuracy is also limited by
 119 finite-particle effects and by choices such as bandwidth selection and early
 120 stopping, which can mute the benefits of increasing time-integration order
 121 at a fixed computational budget. We therefore utilize a first-order multirate
 122 construction using Euler updates that preserves a macro step of size h for
 123 the drift while resolving the repulsion with m substeps:

$$x^{t+1} = \Psi_{\text{drift}}^h \circ \left(\Psi_{\text{rep}}^{h/m} \right)^m (x^t). \quad (7)$$

Algorithm 1 Fixed multirate SVGD (MR-SVGD)

Require: Particles $\{x_i\}_{i=1}^N$, step size h , repulsion substeps m , kernel k

```
1:  $x \leftarrow \{x_i\}$ 
2: for  $s = 1$  to  $m$  do
3:   compute repulsion  $f_{\text{rep}}(x)$  using eq. (5)
4:    $x \leftarrow x + (h/m) f_{\text{rep}}(x)$ 
5: end for
6: compute drift  $f_{\text{drift}}(x)$  using eq. (4)
7:  $x \leftarrow x + h f_{\text{drift}}(x)$ 
8: return  $x$ 
```

124 This retains a coarse drift step while stabilizing the repulsive interaction
125 when particles crowd, at the cost of additional kernel evaluations. The cor-
126 responding implementation-level procedure is summarized in algorithm 1.

127 *Adaptive multirate SVGD (Adapt-MR-SVGD).* Although fixed- m schedules
128 are simple and robust, they impose a uniform drift resolution along the entire
129 trajectory. As a result, they may spend unnecessary computational effort in
130 benign regions while providing insufficient resolution in locally stiff regimes.
131 To address this imbalance, we choose the number of drift microsteps m adap-
132 tively at each macro step, while retaining one repulsion update per macro
133 step. This follows the standard local-error-control perspective used in ODE
134 integrators, where local error estimates are used to adjust resolution [17, Ch.
135 II]. After the repulsion update, denote the state by $x = \Psi_{\text{rep}}^h(x^t)$. We esti-
136 mate the local drift-discretization error by comparing one drift step of size h

137 with two consecutive drift half-steps of size $h/2$,

$$x_{\text{full}} = x + h f_{\text{drift}}(x), \quad (8)$$

$$x_{\text{half}} = x + \frac{h}{2} f_{\text{drift}}(x), \quad (9)$$

$$\tilde{x}_{\text{full}} = x_{\text{half}} + \frac{h}{2} f_{\text{drift}}(x_{\text{half}}). \quad (10)$$

138 We then form the relative discrepancy

$$\rho = \frac{\|\tilde{x}_{\text{full}} - x_{\text{full}}\|_2}{\|x\|_2 + \epsilon}, \quad (11)$$

139 which serves as a dimensionless local error indicator for the drift update.
 140 $\epsilon > 0$ is a small constant to avoid division by zero. For a first-order drift
 141 integrator, this local error behaves as [17, Ch. II.4]:

$$\rho(h) = Ch^2 + \mathcal{O}(h^3),$$

142 where C is the principal error constant dependent on f and its derivatives.
 143 If the drift step is replaced by m microsteps of size h/m , then

$$\rho_m \approx C(h/m)^2 \approx \rho/m^2.$$

144 Imposing the target local error tolerance $\rho_m \approx \text{Tol}$ gives $m \approx \sqrt{\rho/\text{Tol}}$. We
 145 therefore use the clipped integer controller

$$m \leftarrow \text{clip}\left(\left\lceil \sqrt{\rho/\text{Tol}} \right\rceil, m_{\min}, m_{\max}\right), \quad (12)$$

146 which increases m when $\rho > \text{Tol}$ and decreases it when $\rho \leq \text{Tol}$, while
 147 enforcing the robustness bounds $m_{\min} \leq m \leq m_{\max}$. The macro-step update
 148 is then

$$x^{t+1} = \left(\Psi_{\text{drift}}^{h/m}\right)^m \circ \Psi_{\text{rep}}^h(x^t), \quad m = m(x^t, h). \quad (13)$$

149 This strategy increases drift resolution only when the local error indicator
 150 suggests stiffness or rapid variation in the drift component. Algorithm 2
 151 provides the complete algorithm for this method.

152 *Remark 1.* In practice, if $m \leq 2$ we can reuse the already computed, locally-
 153 higher-order solution $\tilde{x}_{\text{full}}$ to avoid extra drift evaluations.

154 3. Experimental Setup

155 We evaluate multirate SVGD variants on six benchmark groups: a 50D
 156 Gaussian target, a suite of 2D synthetic targets, UCI logistic regression, a
 157 2D Gaussian mixture (mix8), a one-hidden-layer Bayesian neural network
 158 (BNN), and a large-scale hierarchical logistic-regression (HLR) benchmark.
 159 The comparison focuses on both statistical quality and computational cost.
 160 This section defines the shared diagnostics and protocol; benchmark-specific
 161 details are introduced where needed below.

162 3.1. Evaluation Metrics

163 Evaluation combines distributional fidelity, predictive performance, mix-
 164 ing, and explicit computational cost accounting. Metrics used across multiple
 165 benchmarks are defined here once; benchmark-specific diagnostics are intro-
 166 duced locally.

167 *Moment fidelity.* When reference moments are available, we report $e_\mu =$
 168 $\|\hat{\mu} - \mu_{\text{ref}}\|_2$ and $e_\Sigma = \|\hat{\Sigma} - \Sigma_{\text{ref}}\|_F$, where $\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$ and $\hat{\Sigma} = \frac{1}{N} \sum_{i=1}^N (x_i -$
 169 $\hat{\mu})(x_i - \hat{\mu})^\top$. These summarize first- and second-order bias, which is critical
 170 in the Gaussian and synthetic-target settings.

171 *Stein discrepancy.* For distributional fidelity, we use kernel Stein discrepancy
 172 (KSD) with score $s_p(x) = \nabla_x \log p(x)$ and an RBF base kernel $k_h(\cdot, \cdot)$ [18, 19].
 173 The corresponding Stein kernel is

$$\begin{aligned} \kappa_p(x, x') &= s_p(x)^\top s_p(x') k_h(x, x') + s_p(x)^\top \nabla_{x'} k_h(x, x') \\ &\quad + s_p(x')^\top \nabla_x k_h(x, x') + \text{tr}(\nabla_x \nabla_{x'} k_h(x, x')) , \end{aligned} \quad (14)$$

174 and the empirical discrepancy is

$$\text{KSD}(X; p) = \left(\frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \kappa_p(x_i, x_j) \right)^{1/2} . \quad (15)$$

175 KSD measures how far the empirical particle distribution is from satisfying
 176 the target’s Stein identity. Intuitively, it acts as a global goodness-of-fit
 177 metric: it is small when the particles collectively match the target law, and
 178 large when they exhibit systematic errors such as bias, incorrect spread, or
 179 missed modes.

180 *Mean log-density fit.* We complement KSD with the empirical mean log-
 181 density

$$\overline{\log p}(X) = \frac{1}{N} \sum_{i=1}^N \log p(x_i), \quad (16)$$

182 which measures how strongly the particles concentrate in high-density regions
 183 of the target. Unlike KSD, this is a pointwise criterion: it rewards samples
 184 for lying in locally plausible regions, but by itself it does not certify global
 185 distributional agreement. Reporting eqs. (15) and (16) therefore separates
 186 global distributional fidelity from local density concentration.

187 *Predictive performance and calibration.* For the binary-classification exper-
 188 iments (UCI and BNN tasks), we report accuracy, negative log-likelihood,

189 and expected calibration error ¹,

$$\text{NLL} = -\frac{1}{M} \sum_{m=1}^M [y_m \log \hat{p}_m + (1 - y_m) \log(1 - \hat{p}_m)], \quad (17)$$

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{M} |\text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b)|. \quad (18)$$

190 In all predictive experiments, ECE is computed with $B = 10$ equal-width
 191 bins on $[0, 1]$. Together, these metrics capture thresholded classification per-
 192 formance, probabilistic predictive fit, and calibration quality.

193 *Mixing efficiency.* To quantify dependence, we report ESS as a supplemen-
 194 tal univariate mixing diagnostic, computed from the first coordinate using
 195 a standard autocorrelation-based estimator with initial-positive truncation
 196 [20]. For particle methods at a checkpoint, the diagnostic sequence is formed
 197 from the first coordinate across particles; for single-chain baselines, it is
 198 formed from a rolling window of recent iterates. Because this quantity is
 199 one-dimensional, we do not interpret ESS as a full multivariate efficiency
 200 measure but as a complementary statistic.

201 *Cost accounting.* Alongside wall-clock time, we track gradient and kernel
 202 evaluation counts to support hardware-agnostic efficiency comparisons across
 203 methods with different workloads.

204 *Remark 2.* For the common metrics above, lower is better for e_μ , e_Σ , KSD,
 205 NLL, ECE, wall-clock time, and gradient- and kernel-evaluation counts, whereas

¹Accuracy as the fraction of correctly classified examples under a 0.5 threshold on the posterior predictive probability. $\text{acc}(\mathcal{B}_b)$ is the empirical accuracy in bin $\mathcal{B}_b$ under the same threshold, and $\text{conf}(\mathcal{B}_b)$ is the mean predictive probability in that bin.

206 higher is better for ESS, accuracy, and mean log-probability.

207 3.2. Experimental Protocol

208 Across benchmarks, we compare the same method family whenever ap-
 209 plicable: vanilla SVGD, Strang-split SVGD, fixed multirate SVGD (MR-
 210 SVGD), adaptive multirate SVGD (Adapt-MR-SVGD), and the single-chain
 211 baselines SGLD and an SGHMC-style method. The SVGD-family updates
 212 are those defined in section 2, namely eqs. (1), (6), (7) and (13).

213 Single-chain baselines use the standard stochastic-gradient update

$$x_{t+1} = x_t + \eta \nabla \log p(x_t) + \sqrt{2\eta} \xi_t, \quad \xi_t \sim \mathcal{N}(0, I), \quad (19)$$

214 for SGLD [1], and

$$v_{t+1} = \beta v_t + \eta \nabla \log p(x_t) + \sqrt{2\alpha\eta} \xi_t, \quad x_{t+1} = x_t + v_{t+1}, \quad (20)$$

215 for an SGHMC-style baseline [2]. Here η is the step size, α is a friction or
 216 noise parameter, and $\beta \in (0, 1)$ is the momentum decay.

217 *Particle configuration and kernel choice.* For particle methods, we use $N =$
 218 128 particles; for single-chain methods, we compute metrics from a rolling
 219 window of recent iterates to obtain comparable finite-sample summaries.
 220 Unless stated otherwise, particle methods use a Gaussian RBF kernel with
 221 bandwidth selected by the median heuristic and a benchmark-specific scal-
 222 ing factor. For Mix8, we instead use a shared multiscale RBF kernel with
 223 bandwidth factors $\{0.5, 1, 2\}$ to better probe local spacing and longer-range
 224 interactions across modes.

225 *Implementation safeguards.* To improve numerical robustness, the multirate
 226 implementations clip raw score components to the interval $[-50, 50]$ before
 227 assembling the update. Benchmarks with pronounced instability additionally
 228 use a short non-finite fail-fast guard, which we note locally when it is active.

229 *Checkpointing and model selection.* All benchmarks use checkpointed eval-
 230 uation at benchmark-specific intervals and a patience-based early-stopping
 231 rule triggered by persistent degradation of a designated monitor metric. Pure
 232 sampling benchmarks use KSD as the monitor (50D Gaussian, 2D synthetic
 233 targets, and Mixture2D), whereas predictive benchmarks use held-out NLL
 234 (UCI logistic regression, BNN, and HLR). For each run, we retain the best
 235 finite checkpoint under that monitor before stopping. Accuracy and ECE
 236 are reported for predictive tasks but are never used as stopping criteria.

237 4. Results

238 In the following, we introduce each benchmark, explain its relevance to
 239 particle-based sampling, and report the corresponding empirical results. The
 240 benchmark suite consists of a 50D Gaussian target, 2D synthetic targets, a
 241 multimodal Gaussian mixture (Mix8), UCI logistic regression, a Bayesian
 242 neural network, and large-scale hierarchical logistic regression.

243 4.1. 50D Gaussian

244 This benchmark tests the stability of sampling in a high-dimensional set-
 245 ting with anisotropic covariance structure. We run 1,000 sampling iterations
 246 with checkpointing every 50 steps and a 5-seed ablation. Reported met-
 247 rics are moment errors ($\|\hat{\mu} - \mu\|_2$ and $\|\hat{\Sigma} - \Sigma\|_F$), ESS, KSD, and time-to

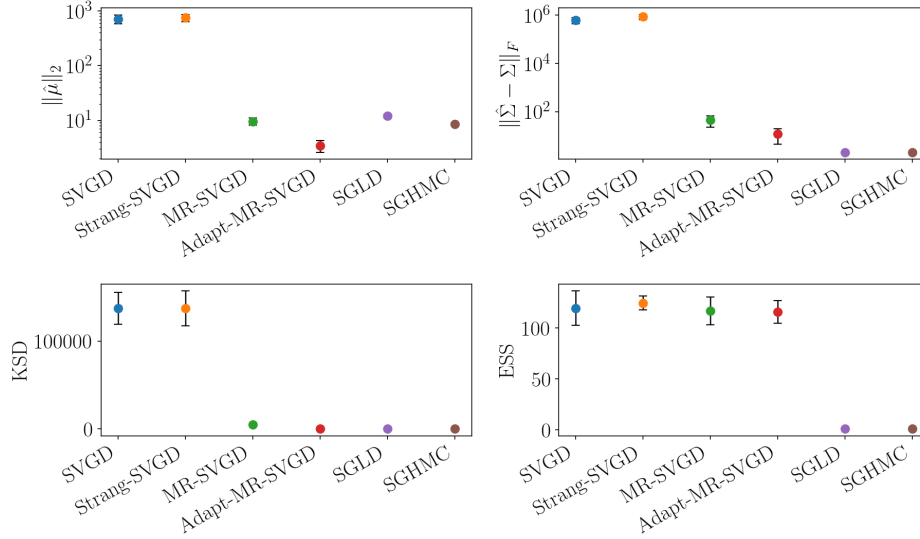


Figure 1: 50D Gaussian final-checkpoint summary across methods. The four panels report $\|\hat{\mu} - \mu\|_2$, $\|\hat{\Sigma} - \Sigma\|_F$, KSD, and ESS, respectively; markers show mean values across seeds and error bars indicate one standard deviation. Lower is better for the first three metrics, whereas higher is better for ESS. This figure highlights robustness differences under strong anisotropy: Adapt-MR-SVGD is the only particle variant that keeps both moment errors and KSD under control.

best-checkpoint.

Among the methods, fig. 1 shows that Adapt-MR-SVGD is the only variant that keeps both moment errors and KSD under control in this anisotropic Gaussian setting: it substantially improves on MR-SVGD and avoids the severe degradation exhibited by vanilla SVGD and Strang-SVGD. The chain baselines remain competitive on some final-checkpoint fidelity metrics, but they do so with ESS near 4, far below the particle methods. Within the particle family, fig. 2 shows that MR-SVGD offers a cheaper but less accurate baseline, whereas Adapt-MR-SVGD achieves the strongest quality at a

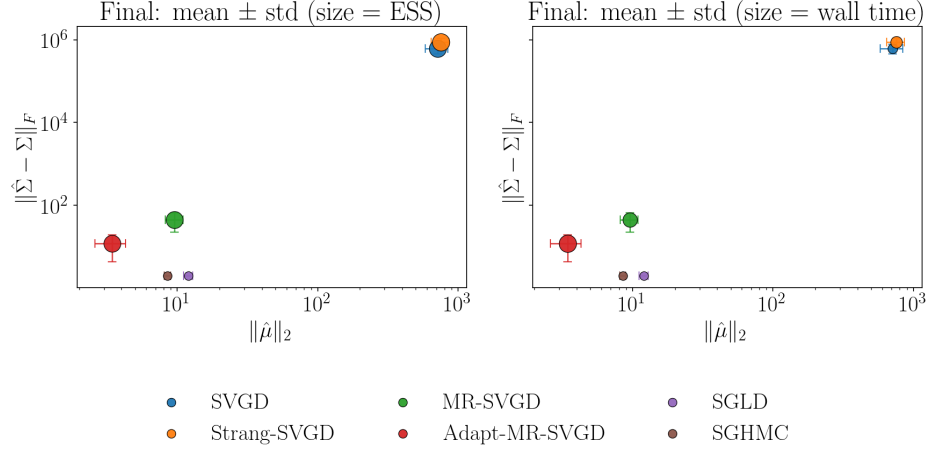
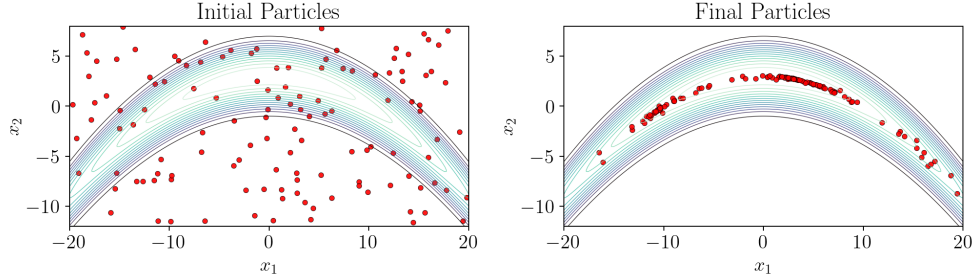


Figure 2: 50D Gaussian: final mean $\pm$ std Pareto plot in moment-error space with marker size encoding ESS (left) and wall time (right).

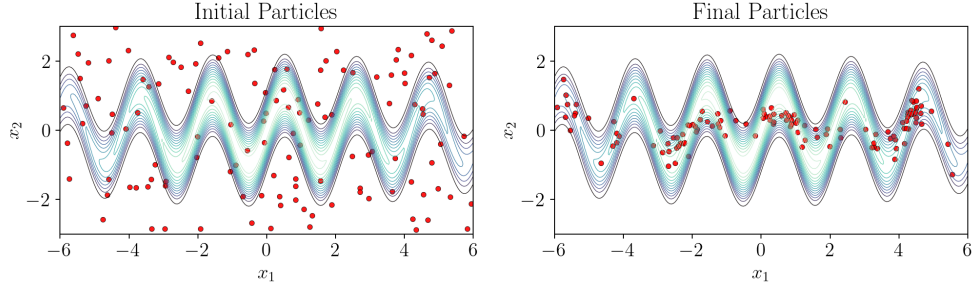
higher computational cost.

4.2. 2D synthetic targets

In this benchmark, we evaluate a suite of 2D targets with varying geometry and multimodality, including banana, ring, squiggly, two-moons, and funnel. To contextualize the geometry of these targets, fig. 3 shows short-run visualization panels for representative cases. The metrics are averaged over five seeds, each with 1,000 iterations and checkpointing every 20 steps. Here KSD serves both as the stop monitor and the primary quality metric. We add a safety guard against non-finite KSD values in the stopping policy, because several targets can produce unstable particle trajectories. We report mean log probability as a secondary density-concentration score, gradient/kernel evaluations and wall time as cost metrics, and ESS as a supplemental mixing proxy. Moment errors and grid-based L_1 discrepancy are retained as diagnostics, but they are not used in the main 2D summary table in table 1.



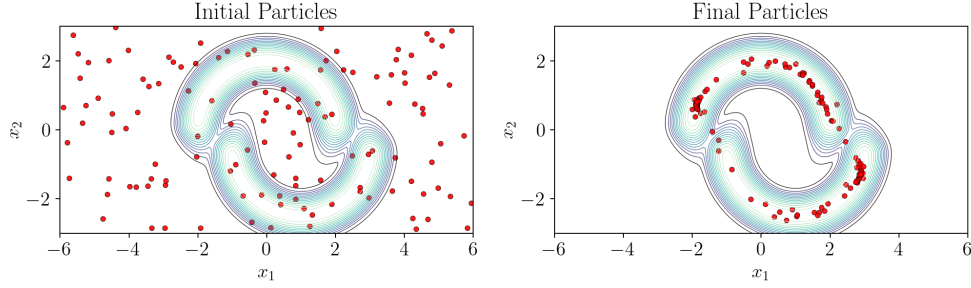
(a) Banana target: initial and short-run final particle states.



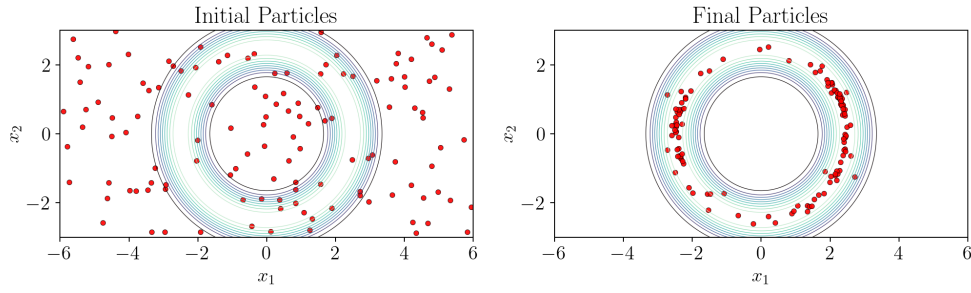
(b) Squiggly target: initial and short-run final particle states.

Figure 3: 2D target visualization panels produced with Adapt-MR-SVGD under visualization-only settings. Each subfigure shows target-density contours with initial particles (left) and short-run final particles (right).

271 Across the full 2D suite, table 1 shows that Adapt-MR-SVGD gives the
 272 strongest overall distributional fidelity: it is the only particle method with
 273 zero failures, attains the best overall median KSD, and also attains the best
 274 median mean log-density while retaining high ESS. The gain is concentrated
 275 on the geometry-driven ring, squiggly, and two-moons targets, where Adapt-
 276 MR-SVGD achieves the best mean KSD over seeds (approximately 0.231,
 277 1.03, and 0.308, respectively) and the single-rate particle methods degrade
 278 sharply. On the easier banana target, by contrast, all particle methods attain
 279 nearly identical KSD values around 0.116, consistent with the comparatively



(c) Two-moons target: initial and short-run final particle states.



(d) Ring target: initial and short-run final particle states.

Figure 3: 2D target visualization panels (continued).

280 mild geometry illustrated in fig. 3; here multirate treatment is not especially
 281 beneficial.

282 The funnel target remains the main exception. Fixed MR-SVGD attains
 283 the best mean KSD on funnel (0.261 versus 0.456 for Adapt-MR-SVGD),
 284 but Adapt-MR-SVGD still remains fully stable and yields substantially bet-
 285 ter mean log-density (-0.023 versus -2.89). Vanilla SVGD and Strang-
 286 SVGD each fail on four of the five funnel seeds, so finite-run medians alone
 287 would understate their instability; this is why table 1 reports failure counts
 288 alongside the quality and cost summaries. Overall, these 2D results sharpen
 289 the main qualitative claim: adaptive multirate integration is most valuable
 290 when the interaction between drift and repulsion becomes strongly geometry-

Method	Fails↓	KSD↓	mean logp↑	ESS↑	grad evals↓	wall (s)↓
Adapt-MR-SVGD	0	0.308	-0.175	128.0	182	0.79
MR-SVGD	0	40.13	-278.0	121.8	80	1.13
SVGD	4	26.02	-1.93×10^4	128.0	80	0.32
Strang-SVGD	4	30.09	-1.20×10^5	121.7	121	1.55

Table 1: 2D targets quality-cost summary across particle methods. Entries are medians over the 25 target-seed runs, using the best checkpoint selected by the KSD monitor for each run. “Fails” counts runs whose KSD monitor never attained a finite value; methods are ordered first by failure count and then by median KSD. KSD is the primary distributional quality metric; mean log probability is reported as a secondary score, ESS as a supplemental mixing indicator, and gradient evaluations / wall time as cost metrics. Moment errors and grid L_1 remain diagnostic quantities and are omitted from this main summary. The main pattern is that Adapt-MR-SVGD is the only zero-failure particle method and gives the best overall KSD across the suite, while banana shows little separation and funnel remains a nuanced exception.

291 dependent, rather than as a uniform improvement on every benign target. In
292 this setting, ESS is much less discriminative than KSD and mean log-density,
293 so we treat it as a supplemental mixing indicator rather than a primary qual-
294 ity metric.

295 4.3. *Mixture2D (mix8)*

296 This benchmark emphasizes multimodal exploration and mass allocation
297 across modes. All methods start from the same small Gaussian centered at
298 the origin, away from the modal ring, and then run for a fixed budget of
299 1,000 iterations with checkpointing every 20 steps across 5 seeds. For this
300 benchmark we use a shared multiscale RBF kernel with bandwidth factors
301 $\{0.5, 1, 2\}$ for all particle methods so that local spacing and longer-range

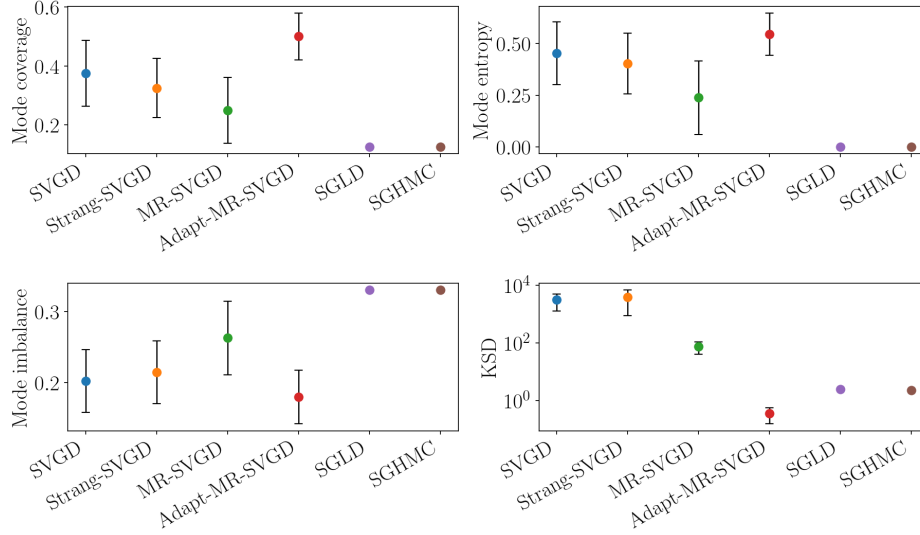


Figure 4: Mixture2D (mix8) fixed-budget final-checkpoint summary across methods. The four panels report mode coverage, mode entropy, mode imbalance, and KSD; markers show mean values across seeds and error bars indicate one standard deviation. Higher is better for coverage and entropy, while lower is better for mode imbalance and KSD.

302 repulsion are both represented on the modal ring. We treat this as an
 303 experimental kernel-choice ablation rather than a change to the multirate
 304 update itself; prior SVGD work has also considered broader kernel families
 305 beyond a single scalar RBF [5, 21]. Let p_k denote the empirical mass as-
 306 signed to mixture center k by nearest-center assignment. We define mode
 307 coverage as the fraction of modes with $p_k \geq 0.05$, normalized mode entropy
 308 as $-\sum_{k=1}^K p_k \log p_k / \log K$, and mode imbalance as the standard deviation of
 309 $(p_1, \dots, p_K)$. The primary metrics are therefore mode coverage, normalized
 310 mode entropy, and mode imbalance; KSD and mean log probability are re-
 311 tained as secondary diagnostics, with ESS reported as a supplemental mixing
 312 proxy.

Table 2: Mixture2D (mix8) mean final metrics over five seeds at the fixed 1,000-iteration budget. Higher is better for coverage and entropy; lower is better for KSD. We omit minimum mass because it is zero for all methods at this budget. This table summarizes the fixed-budget multimodal result: Adapt-MR-SVGD gives the strongest overall exploration, while vanilla and Strang spread more than MR-SVGD but remain far less stable by KSD.

Method	Coverage $\uparrow$	Entropy $\uparrow$	KSD $\downarrow$
Adapt-MR-SVGD	0.500	0.544	0.360
SVGD	0.375	0.453	3162.9
Strang-SVGD	0.325	0.403	3883.0
MR-SVGD	0.250	0.238	74.3
SGHMC	0.125	0.000	2.29
SGLD	0.125	0.000	2.53

313 Under this harder off-mode initialization, adaptive MR-SVGD still gives
 314 the strongest overall multimodal exploration at the fixed budget, now reach-
 315 ing the highest mean final mode coverage and mode entropy across seeds
 316 (0.500 and 0.544, respectively); table 2 provides the corresponding mean-
 317 over-seeds summary, while fig. 4 carries the cross-seed variability. The shared
 318 multiscale kernel also makes the single-rate particle baselines more compet-
 319 itive than under the plain single-RBF setting: vanilla SVGD and Strang-
 320 SVGD both cover more modes and attain higher mode entropy, but their
 321 KSD values remain far worse than the adaptive method, so the extra spread
 322 still comes with much weaker stability. Fixed MR-SVGD remains stable and
 323 slightly improves its KSD over the single-RBF version, but it does not match
 324 the adaptive variant on the main multimodal metrics. The imbalance panel

in fig. 4 complements coverage and entropy by showing that the adaptive method also allocates mass more evenly across the discovered modes. SGLD and SGHMC remain trapped in a single mode, with final coverage fixed at 0.125 and zero mode entropy. The minimum mass statistic is still zero for all methods at this budget, so in this redesigned benchmark coverage, entropy, and imbalance are the informative multimodal diagnostics while KSD remains a secondary stability measure.

4.4. UCI logistic regression

We benchmark Bayesian logistic regression on breast_cancer, ionosphere, spambase, and a5a, using a Gaussian prior on weights (as implemented in the benchmark script) and 5 seeds per dataset. We report test accuracy, NLL, ECE, ESS, and mean log probability; KSD is optional and disabled by default in this benchmark configuration. Results are summarized with per-dataset panels in fig. 5 and a grouped best-NLL summary table (particle methods only) in table 3; both use the best finite-NLL checkpoint selected under the global predictive stopping policy [22].

UCI logistic regression serves mainly as a predictive sanity check rather than as the flagship evidence for multirate gains. The rankings are mixed. Adapt-MR-SVGD gives the best overall NLL on breast_cancer, while on ionosphere, spambase, and a5a either SGHMC or a single-rate particle method is competitive or superior. This weaker separation is consistent with the main claim of the paper: multirate benefits are most pronounced on anisotropic, multimodal, or hierarchical posteriors, whereas lower-dimensional logistic regression tasks show smaller and less uniform gains. On the smaller datasets, accuracy is also tightly quantized, so NLL is the more discriminative sum-

350 mary.

351 4.5. Bayesian neural network

352 We evaluate a one-hidden-layer BNN (hidden width 32) on spambase
353 and a5a, using a Gaussian prior, a fixed split seed, and 5 sampler seeds per
354 dataset. For each dataset, we carve a validation split from the training data,
355 select checkpoints by validation NLL under the global predictive stopping
356 policy, and report test accuracy, test NLL, test ECE, ESS, and mean log
357 probability at the selected checkpoint. KSD is optional and disabled by
358 default in this configuration.

359 The two BNN datasets give a more informative but less uniform predictive
360 story than the earlier small-data configuration. On spambase, Adapt-MR-
361 SVGD attains the best overall test NLL (0.174) and the highest mean test
362 accuracy (0.940), with the remaining particle methods clustered together
363 around 0.187. On a5a, the ranking shifts: MR-SVGD attains the best overall
364 test NLL (0.334) and the highest mean test accuracy (0.844), Adapt-MR-
365 SVGD remains the second-best particle method (0.347), and the single-rate
366 particle variants are again worse. The chain baselines are competitive in
367 isolated cases—SGHMC on spambase and SGLD on a5a—but the multirate
368 particle methods form the strongest pair overall across the two datasets.
369 This makes the BNN benchmark corroborative rather than primary evidence:
370 multirate structure improves predictive quality over the single-rate particle
371 baselines, but the preferred multirate variant depends on the dataset.

372 4.6. Large-scale hierarchical logistic regression

373 To stress-test multirate behavior in a high-dimensional anisotropic pos-
 374 terior, we use a large-scale hierarchical logistic regression benchmark with
 375 grouped random effects. The model combines high parameter dimension
 376 with funnel-like geometry induced by a global scale parameter, making it a
 377 natural benchmark for joint quality-cost assessment of SVGD variants.

378 *Problem definition.* Given binary outcomes $y_i \in \{0, 1\}$, sparse features $x_i \in$
 379 $\mathbb{R}^p$, and group indices $g_i \in \{1, \dots, G\}$, we define

$$\begin{aligned} y_i \mid x_i, g_i, \beta, \alpha, u &\sim \text{Bernoulli}(\sigma(\alpha + x_i^\top \beta + u_{g_i})), \\ i &= 1, \dots, n, \end{aligned} \quad (21)$$

380 where $\sigma(\cdot)$ is the logistic link. We use a non-centered parameterization for
 381 group effects,

$$\begin{aligned} \beta &\sim \mathcal{N}(0, \sigma_\beta^2 I_p), \quad \alpha \sim \mathcal{N}(0, \sigma_\alpha^2), \\ u_j &= \tau z_j, \quad z_j \sim \mathcal{N}(0, 1), \\ \log \tau &\sim \mathcal{N}(\mu_\tau, \sigma_\tau^2), \end{aligned} \quad (22)$$

382 for $j = 1, \dots, G$. The posterior (up to normalization) is

$$\begin{aligned} \pi(\theta) &\propto \left[\prod_{i=1}^n \text{Bernoulli}(y_i; \sigma(\alpha + x_i^\top \beta + u_{g_i})) \right] \\ &\quad \times p(\beta) p(\alpha) p(z) p(\log \tau), \\ \theta &= (\beta, \alpha, z, \log \tau). \end{aligned} \quad (23)$$

383 *Protocol.* For the long-tail grouped setting, we run $n = 10^6$ observations
 384 with $p = 300$ features and $G = 50,000$ groups. Each method runs up to
 385 1,000 iterations with checkpointing every 20 iterations and 5 random seeds;

particle methods use 32 particles, and single-chain baselines use a rolling-window evaluation set. This benchmark follows the global predictive NLL-based stopping policy and adds a two-checkpoint non-finite fail-fast guard because unstable methods can quickly leave the finite-NLL regime. ECE is reported as a metric but is not used as a stopping trigger.

Result summary. table 5 shows that Adapt-MR-SVGD achieves the best mean best-NLL while remaining finite across all seeds. MR-SVGD is a close second in NLL and reaches its best checkpoint faster on average (92.5s versus 123.3s). SVGD/Strang-SVGD and SGHMC show frequent non-finite behavior in this regime, while SGLD remains finite more often but with substantially worse NLL.

Uniform-group sensitivity check. table 6 provides a compact robustness check under uniform group assignments. Adapt-MR-SVGD remains best in mean best-NLL with full finite-seed coverage (5/5), and MR-SVGD remains close while being fastest among methods with full finite coverage. This supports the same ordering seen in the long-tail regime while keeping the long-tail setting as the primary stress test for our main claim.

5. Conclusions

This work studies a central numerical weakness of SVGD: a single global step size must simultaneously resolve attractive drift and diversity-preserving repulsion even when these components evolve on different effective time scales. We address this mismatch by recasting SVGD as a split interacting-particle flow and integrating the drift and repulsion terms with multirate updates.

410 Across a broad benchmark suite, the main empirical pattern is that multi-
411 rate SVGD improves robustness and quality-cost tradeoffs relative to vanilla
412 SVGD, with the strongest gains on anisotropic, multimodal, and hierarchi-
413 cal targets. Fixed multirate schedules provide a simpler robustness baseline,
414 while adaptive drift resolution offers the strongest performance when local
415 stiffness changes along the trajectory. At the same time, no single vari-
416 ant dominates every metric and every regime, reinforcing the importance of
417 evaluating quality and cost jointly rather than through a single summary
418 statistic.

419 Several limitations remain. Current adaptivity is centered on drift reso-
420 lution, while repulsion frequency is still controlled by simple schedules; ESS
421 remains a univariate proxy; and dense kernel interactions continue to con-
422 strain scaling at large particle counts. These observations suggest clear
423 next steps: tighter adaptive controllers that coordinate drift and repul-
424 sion updates, approximate or structured kernel computations for large- N
425 regimes, and stronger theoretical links between multirate stability analysis
426 and particle-based Bayesian inference guarantees. Together, these directions
427 can move multirate SVGD from a robust empirical strategy toward a more
428 principled and scalable foundation for modern Bayesian computation.

429 **Data availability**

430 Code, experiment scripts, and manuscript sources for this study are main-
431 tained in a public GitHub repository.² Public benchmark inputs used in the

²<https://github.com/csml-beach/multirate-sampling>

432 predictive experiments are derived from standard UCI datasets; all other
433 benchmark targets are synthetic and are generated directly by the repository
434 code. Generated figures, metrics, and supplementary animation artifacts are
435 available in the project repository for reference and reproduction.

436 **Declaration of generative AI and AI-assisted technologies**

437 The jax codebase for this project was developed in part with the assistance
438 of OpenAI Codex. After using this tool/service, the author reviewed and
439 edited the content as needed and takes full responsibility for the content of
440 the published article.

441 **References**

- 442 [1] M. Welling, Y. W. Teh, Bayesian learning via stochastic gradient
443 langevin dynamics, in: International Conference on Machine Learning,
444 2011.
- 445 [2] T. Chen, E. Fox, C. Guestrin, Stochastic gradient hamiltonian monte
446 carlo, in: International Conference on Machine Learning, 2014.
- 447 [3] C. Liu, J. Zhuo, P. Cheng, R. Zhang, J. Zhu, Understanding and
448 accelerating particle-based variational inference, in: K. Chaudhuri,
449 R. Salakhutdinov (Eds.), Proceedings of the 36th International Confer-
450 ence on Machine Learning, Vol. 97 of Proceedings of Machine Learning
451 Research, PMLR, 2019, pp. 4082–4092.
452 URL <https://proceedings.mlr.press/v97/liu19i.html>

- 453 [4] J. Zhuo, C. Liu, J. Shi, J. Zhu, N. Chen, B. Zhang, Message passing stein
454 variational gradient descent, in: J. Dy, A. Krause (Eds.), Proceedings
455 of the 35th International Conference on Machine Learning, Vol. 80 of
456 Proceedings of Machine Learning Research, PMLR, 2018, pp. 6018–
457 6027.
458 URL <https://proceedings.mlr.press/v80/zhuo18a.html>
- 459 [5] D. Wang, Z. Tang, C. Bajaj, Q. Liu, Stein variational gradient descent
460 with matrix-valued kernels, in: Advances in Neural Information Pro-
461 cessing Systems, 2019.
- 462 [6] A. N. Subrahmanya, A. A. Popov, A. Sandu, Ensem-
463 ble variational fokker-planck methods for data assimila-
464 tion, Journal of Computational Physics 523 (2025) 113681.
465 doi:<https://doi.org/10.1016/j.jcp.2024.113681>.
- 466 [7] Q. Liu, D. Wang, Stein variational gradient descent: A general pur-
467 pose bayesian inference algorithm, in: Advances in Neural Information
468 Processing Systems, 2016.
- 469 [8] J. Zhang, R. Zhang, L. Carin, C. Chen, Stochastic particle-optimization
470 sampling and the non-asymptotic convergence theory, in: S. Chiappa,
471 R. Calandra (Eds.), Proceedings of the Twenty Third International Con-
472 ference on Artificial Intelligence and Statistics, Vol. 108 of Proceedings
473 of Machine Learning Research, PMLR, 2020, pp. 1877–1887.
474 URL <https://proceedings.mlr.press/v108/zhang20d.html>
- 475 [9] L. Pareschi, G. Russo, Implicit-explicit Runge-Kutta schemes for stiff

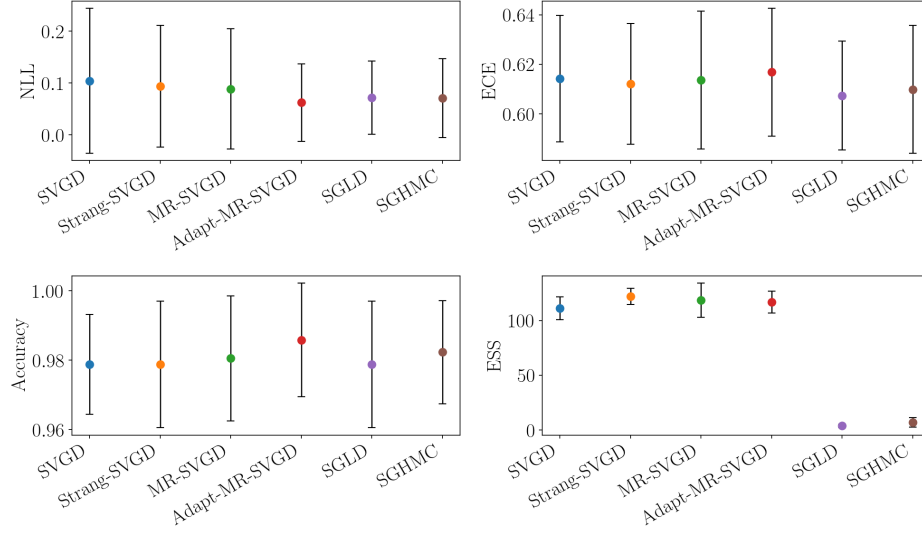
- 476 systems of differential equations, in: Recent trends in numerical analysis,
477 Nova Science Publishers, Inc., 2000, pp. 269–288.
- 478 [10] A. Sarshar, S. Roberts, A. Sandu, Parallel implicit-explicit general linear
479 methods, Communications on Applied Mathematics and Computation
480 3 (2021) 649–669. doi:10.1007/s42967-020-00083-5.
- 481 [11] M. Günther, A. Sandu, Multirate generalized additive Runge-
482 Kutta methods, Numerische Mathematik 133 (3) (2016) 497–524.
483 doi:10.1007/s00211-015-0756-z.
- 484 [12] A. Sarshar, S. Roberts, A. Sandu, Design of high-order decoupled multi-
485 rate gark schemes, SIAM Journal on Scientific Computing 41 (2) (2019)
486 A816–A847. doi:10.1137/18M1182875.
- 487 [13] A. Sandu, M. Günther, S. Roberts, A. Sarshar, Implicit multi-
488 rate gark methods, Journal of Scientific Computing 87 (2021) 1–32.
489 doi:10.1007/s10915-020-01400-z.
- 490 [14] A. Sarshar, S. Roberts, A. Sandu, Alternating directions im-
491 plicit integration in a general linear method framework, Journal
492 of Computational and Applied Mathematics 387 (2021) 112496.
493 doi:10.1016/j.cam.2019.112496.
- 494 [15] A. Sarshar, S. Roberts, A. Sandu, A fast time-stepping strat-
495 egy for dynamical systems equipped with a surrogate model,
496 SIAM Journal on Scientific Computing 44 (3) (2022) A1405–A1430.
497 doi:10.1137/20M1386281.

- 498 [16] G. Strang, On the construction and comparison of difference schemes,
499 SIAM Journal on Numerical Analysis 5 (3) (1968) 506–517.
- 500 [17] E. Hairer, S. Norsett, G. Wanner, Solving ordinary differential equations
501 I: Nonstiff problems, no. 8 in Springer Series in Computational Mathe-
502 matics, Springer-Verlag Berlin Heidelberg, 1993. doi:10.1007/978-3-540-
503 78862-1.
- 504 [18] Q. Liu, J. Lee, M. Jordan, A kernelized stein discrepancy for goodness-
505 of-fit tests, in: International Conference on Machine Learning, 2016.
- 506 [19] J. Gorham, L. Mackey, Measuring sample quality with kernels, in:
507 D. Precup, Y. W. Teh (Eds.), Proceedings of the 34th International
508 Conference on Machine Learning, Vol. 70 of Proceedings of Machine
509 Learning Research, PMLR, 2017, pp. 1292–1301.
510 URL <https://proceedings.mlr.press/v70/gorham17a.html>
- 511 [20] C. J. Geyer, Practical markov chain monte carlo, Statistical Science 7 (4)
512 (1992) 473–483. doi:10.1214/ss/1177011137.
- 513 [21] A. S. Stordal, R. J. Moraes, P. N. Raanes, G. Evensen, p-kernel stein
514 variational gradient descent for data assimilation and history matching,
515 Mathematical Geosciences 53 (2021) 375–393. doi:10.1007/s11004-021-
516 09937-x.
- 517 [22] D. Dua, C. Graff, Uci machine learning repository, [https://archive.](https://archive.ics.uci.edu/ml)
518 [ics.uci.edu/ml](https://archive.ics.uci.edu/ml), accessed 2025-01-01 (2019).

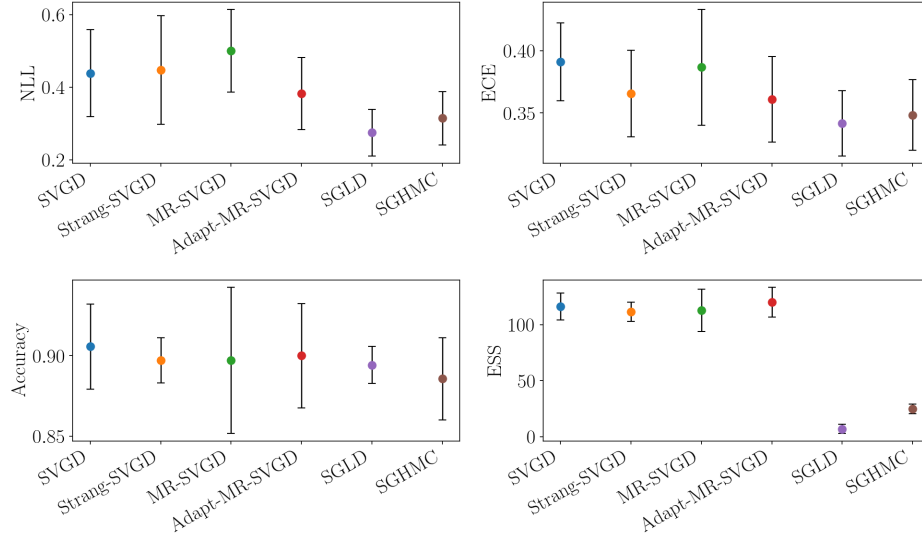
Algorithm 2 Adaptive multirate SVGD (Adapt-MR-SVGD)

Require: Particles $\{x_i\}_{i=1}^N$, step size h , bounds $m_{\min}, m_{\max}$, tolerance Tol

```
1:  $x \leftarrow \{x_i\}$ 
2: compute repulsion  $f_{\text{rep}}(x)$  using eq. (5)
3:  $x \leftarrow x + h f_{\text{rep}}(x)$ 
4: compute drift  $f_{\text{drift}}(x)$  using eq. (4)
5:  $x_{\text{full}} \leftarrow x + h f_{\text{drift}}(x)$ 
6:  $x_{\text{half}} \leftarrow x + (h/2) f_{\text{drift}}(x)$ 
7: compute drift  $f_{\text{drift}}(x_{\text{half}})$ 
8:  $\tilde{x}_{\text{full}} \leftarrow x_{\text{half}} + (h/2) f_{\text{drift}}(x_{\text{half}})$ 
9: estimate relative error  $\rho = \|\tilde{x}_{\text{full}} - x_{\text{full}}\|/(\|x\| + \epsilon)$ 
10:  $m \leftarrow \text{clip}(\lceil \sqrt{\rho/\text{Tol}} \rceil, m_{\min}, m_{\max})$ 
11: if  $m \leq 2$  then
12:      $x \leftarrow \tilde{x}_{\text{full}}$   $\triangleright$  reuse current predictor if accurate
13: else
14:     for  $s = 1$  to  $m$  do
15:         compute drift  $f_{\text{drift}}(x)$ 
16:          $x \leftarrow x + (h/m) f_{\text{drift}}(x)$ 
17:     end for
18: end if
19: return  $x$ 
```

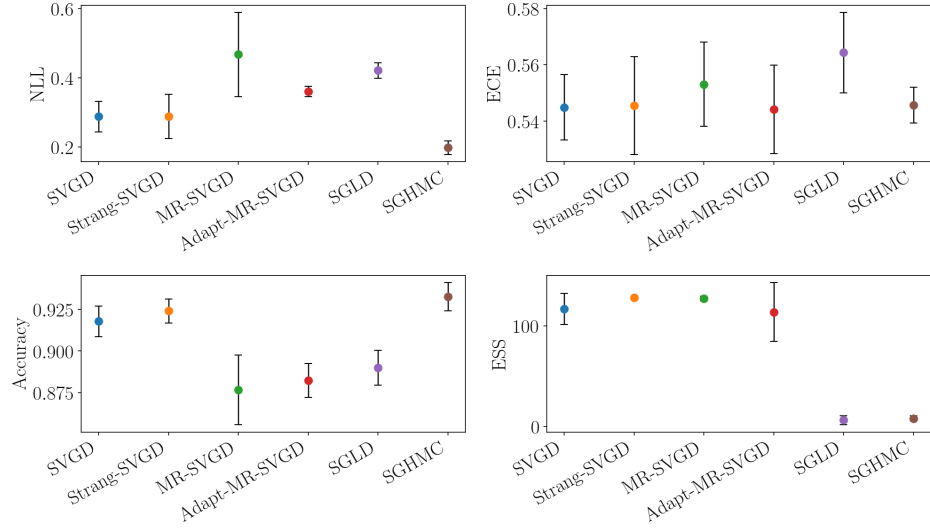


(a) breast_cancer

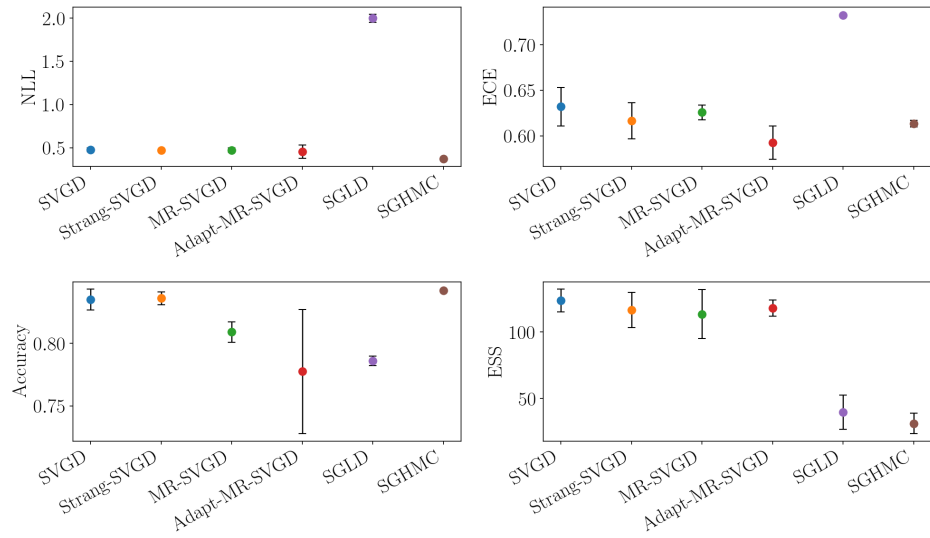


(b) ionosphere

Figure 5: UCI logistic regression summary across datasets. Each subpanel reports the best-checkpoint mean across seeds with one standard deviation for test accuracy, NLL, ECE, and ESS, where the checkpoint is selected by the NLL-based early-stop monitor. Higher is better for accuracy and ESS; lower is better for NLL and ECE. Remaining datasets are shown in the continued panel.



(c) spambase

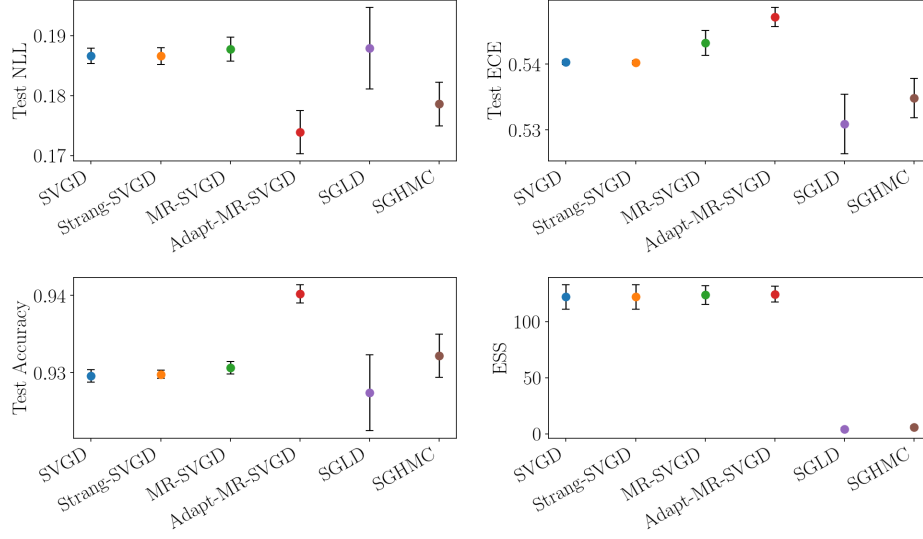


(d) a5a

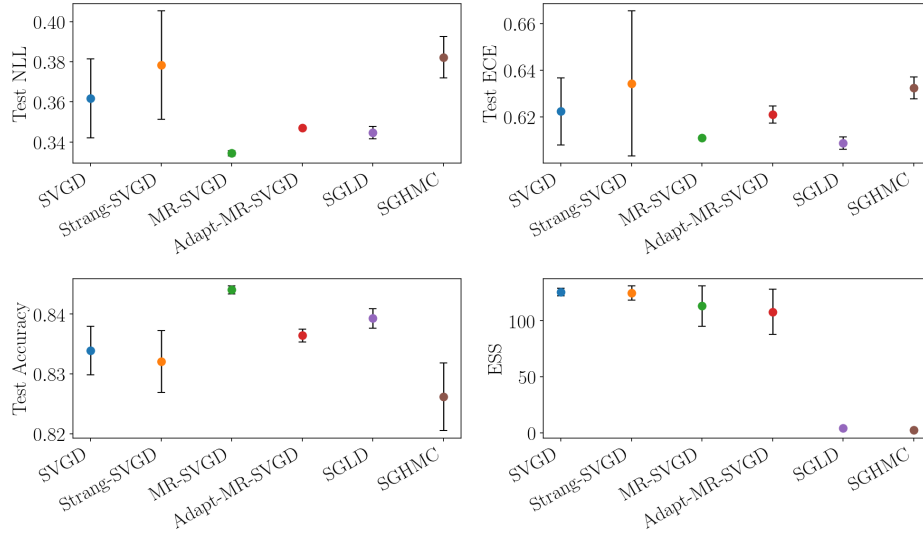
Figure 5: UCI logistic regression summary across datasets (continued).

Method	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
<i>breast_cancer</i>			
Adapt-MR-SVGD	0.062	0.986	8.9
MR-SVGD	0.089	0.981	6.2
Strang-SVGD	0.094	0.979	6.5
SVGD	0.104	0.979	2.0
<i>ionosphere</i>			
Adapt-MR-SVGD	0.383	0.900	3.4
SVGD	0.438	0.906	0.9
Strang-SVGD	0.447	0.897	5.2
MR-SVGD	0.500	0.897	6.1
<i>spambase</i>			
SVGD	0.288	0.918	1.3
Strang-SVGD	0.288	0.924	8.8
Adapt-MR-SVGD	0.361	0.882	8.1
MR-SVGD	0.467	0.877	4.3
<i>a5a</i>			
Adapt-MR-SVGD	0.453	0.778	36.5
Strang-SVGD	0.471	0.836	3.0
MR-SVGD	0.474	0.809	14.6
SVGD	0.478	0.835	2.6

Table 3: UCI logistic-regression particle-method summary over five seeds. For each dataset and method, we select the best finite-NLL checkpoint per seed and report the method-wise mean of that best NLL, the accuracy at that checkpoint, and the corresponding wall time. ECE is summarized in the per-dataset panels of fig. 5.



(a) spambase



(b) a5a

Figure 6: BNN summary across datasets. Each subpanel reports the best-checkpoint mean across seeds with one standard deviation for test accuracy, NLL, ECE, and ESS, where the checkpoint is selected by the validation-NLL early-stop monitor. Higher is better for accuracy and ESS; lower is better for NLL and ECE.

Method	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
<i>spambase</i>			
Adapt-MR-SVGD	0.174	0.940	35.7
Strang-SVGD	0.187	0.930	51.2
SVGD	0.187	0.930	21.1
MR-SVGD	0.188	0.931	51.5
<i>a5a</i>			
MR-SVGD	0.334	0.844	5.3
Adapt-MR-SVGD	0.347	0.836	8.1
SVGD	0.362	0.834	17.5
Strang-SVGD	0.378	0.832	49.6

Table 4: BNN particle-method summary over five seeds. For each dataset and method, we select the best finite-validation-NLL checkpoint per seed and report the method-wise mean of the corresponding test NLL, test accuracy, and wall time. ECE is summarized in the per-dataset panels of fig. 6.

Method	Finite-best↑	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
Adapt-MR-SVGD	5/5	0.613	0.658	123.3
MR-SVGD	5/5	0.623	0.656	92.5
SGHMC	2/5	0.860	0.560	77.9
SVGD	3/5	0.885	0.577	154.2
Strang-SVGD	3/5	1.263	0.579	163.6
SGLD	4/5	1.823	0.576	101.6

Table 5: HLR long-tail summary over five seeds. For each seed and method, we select the best finite-NLL checkpoint; the table reports method-wise means of that best NLL, the accuracy at that checkpoint, and the corresponding wall time. “Seeds with finite best” counts runs that reached at least one finite-NLL checkpoint before stopping. This table is the main hierarchical stress test: Adapt-MR-SVGD achieves the best best-NLL while remaining finite on all seeds, and MR-SVGD is the closest lower-cost baseline.

Method	Finite-best $\uparrow$	Best NLL $\downarrow$	Acc@best NLL $\uparrow$	Wall@best (s) $\downarrow$
Adapt-MR-SVGD	5/5	0.681	0.601	94.6
SVGD	1/5	0.700	0.548	54.9
MR-SVGD	5/5	0.706	0.597	44.2
Strang-SVGD	1/5	0.708	0.584	241.7
SGLD	5/5	0.743	0.528	66.0
SGHMC	5/5	0.791	0.554	79.7

Table 6: HLR uniform-group summary over five seeds. As in table 5, each entry uses the best finite-NLL checkpoint per seed and reports method-wise means for NLL, accuracy at that checkpoint, and wall time.